

Using Forum and Assessment Analytics to Target Student Support Needs

Engineering Dynamics case study, 2022–2024
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Core message: routine course records can identify where large-course students need extra support before misconceptions become terminal exam failures.

-14.0%
2023 late-topic gap

+17.4 pts
2024 gap improvement

98
2024 instructor follow-ups

41.5%
2024 final PC1 variance

477
Vibrations views

92.3%
top-two staff share

1. Why this matters

- Problem: late Dynamics topics produce persistent misconceptions.
- Signal: marks, Piazza threads, follow-ups, and video use reveal where students are stuck.
- Contribution: a reusable workflow for triaging scarce support in large cohorts.

2. Analytics workflow

Question-level marks	Topic means, Q4–Q5 outcomes, and within-student late-topic gap Δ_{late}
Forum logs	Late-topic thread volume, follow-up ratio, and multi-turn dialogue depth
Video analytics	Seasonal use of short conceptual summary resources
SVD/PCA	Within-year loadings and PC1 variance to flag structural bottlenecks

3. Cohorts and constraints

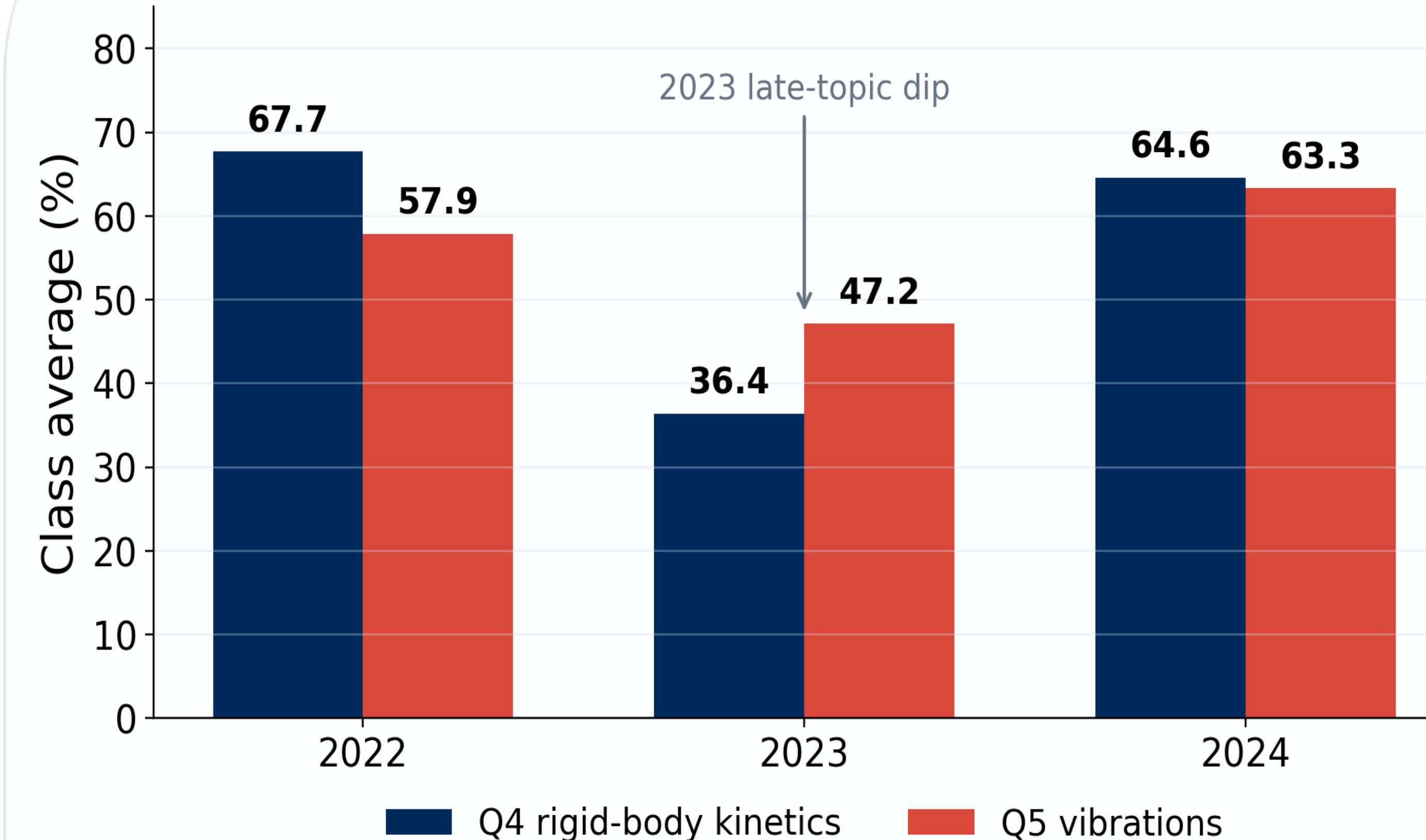
Year	N	Role in case
2022	549	baseline
2023	552	late-topic dip
2024	617	rebound + deeper dialogue

Validity limits

- Observational, instructor-led case study: hypothesis-generating, not causal proof.
- Common exam structure, but item details varied; no psychometric equating was performed.
- Next step: blinded re-grading, independent thread coding, and prior-performance covariates.

4. Key result: the 2023 dip and 2024 rebound

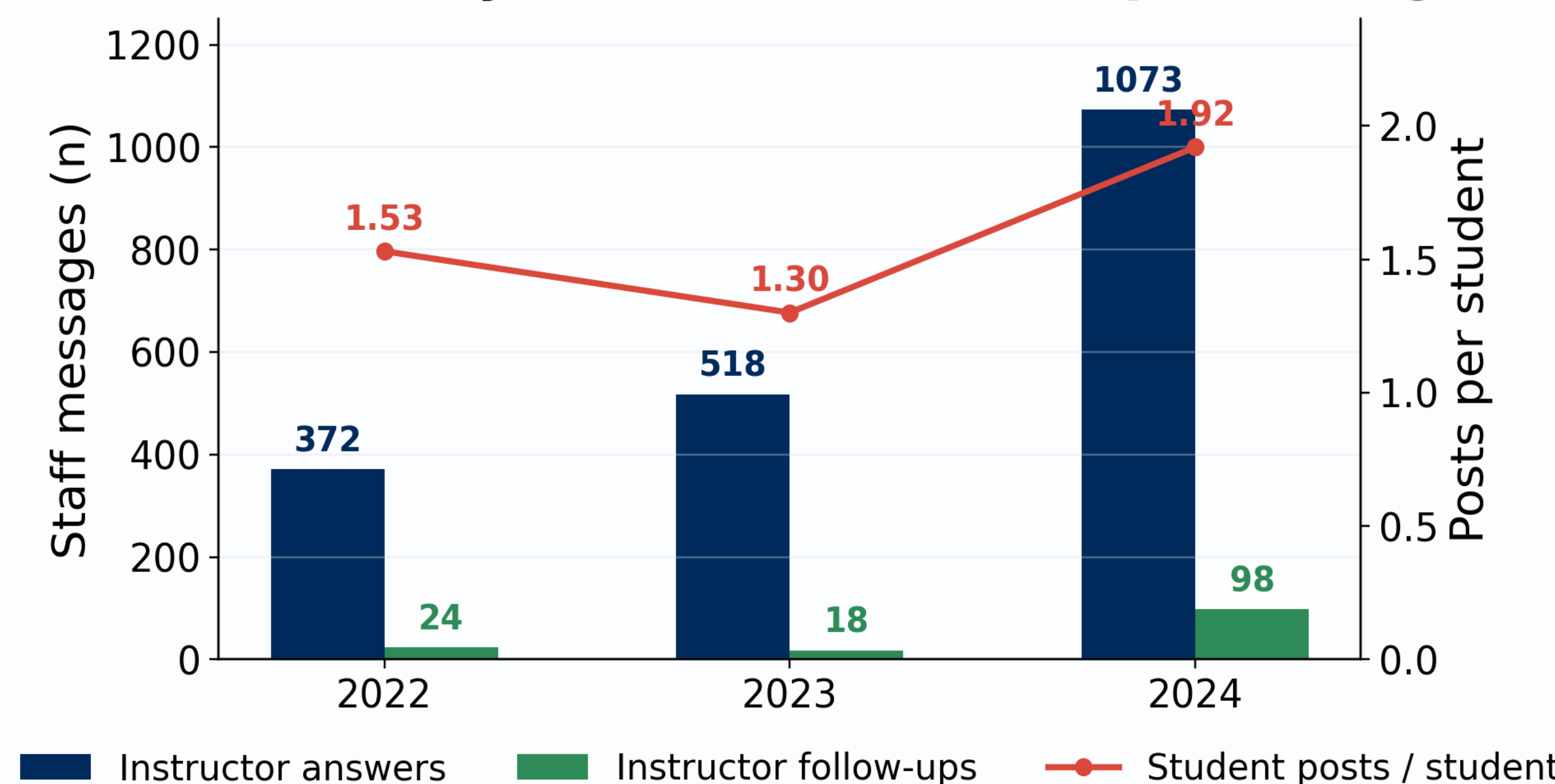
Late-topic performance recovered in 2024



Recovery was concentrated in late-course gatekeeper modules: Q4 +28.2 points and Q5 +16.1 points from 2023 to 2024.

5. Engagement: passive exposure was not enough

Forum activity shifted toward deeper dialogue

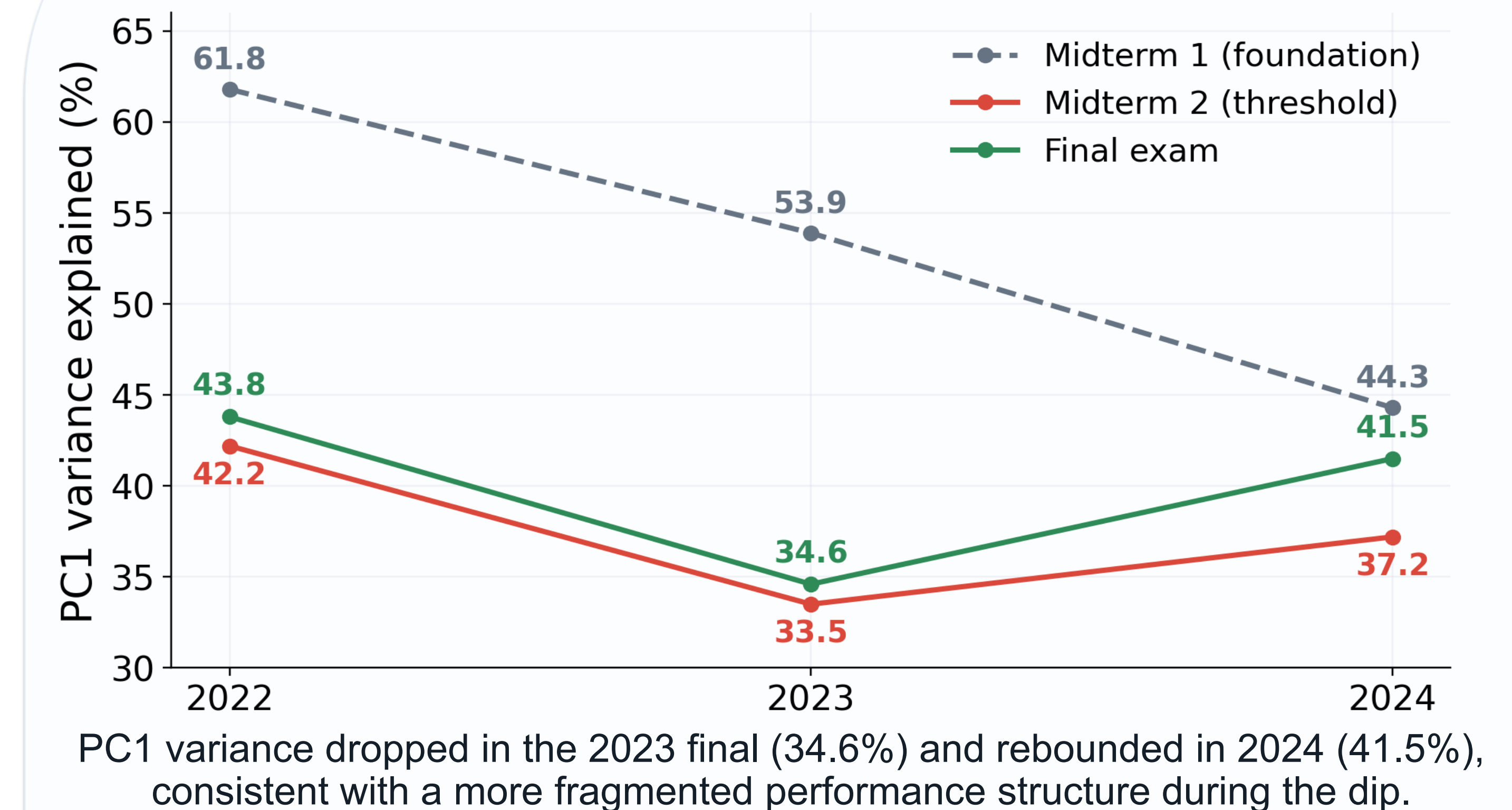


Interpretation

In 2023, students spent more days online but posted less and received fewer follow-ups. In 2024, more student posts and staff follow-ups co-occurred with stronger late-topic outcomes.

6. SVD/PCA: structural bottlenecks

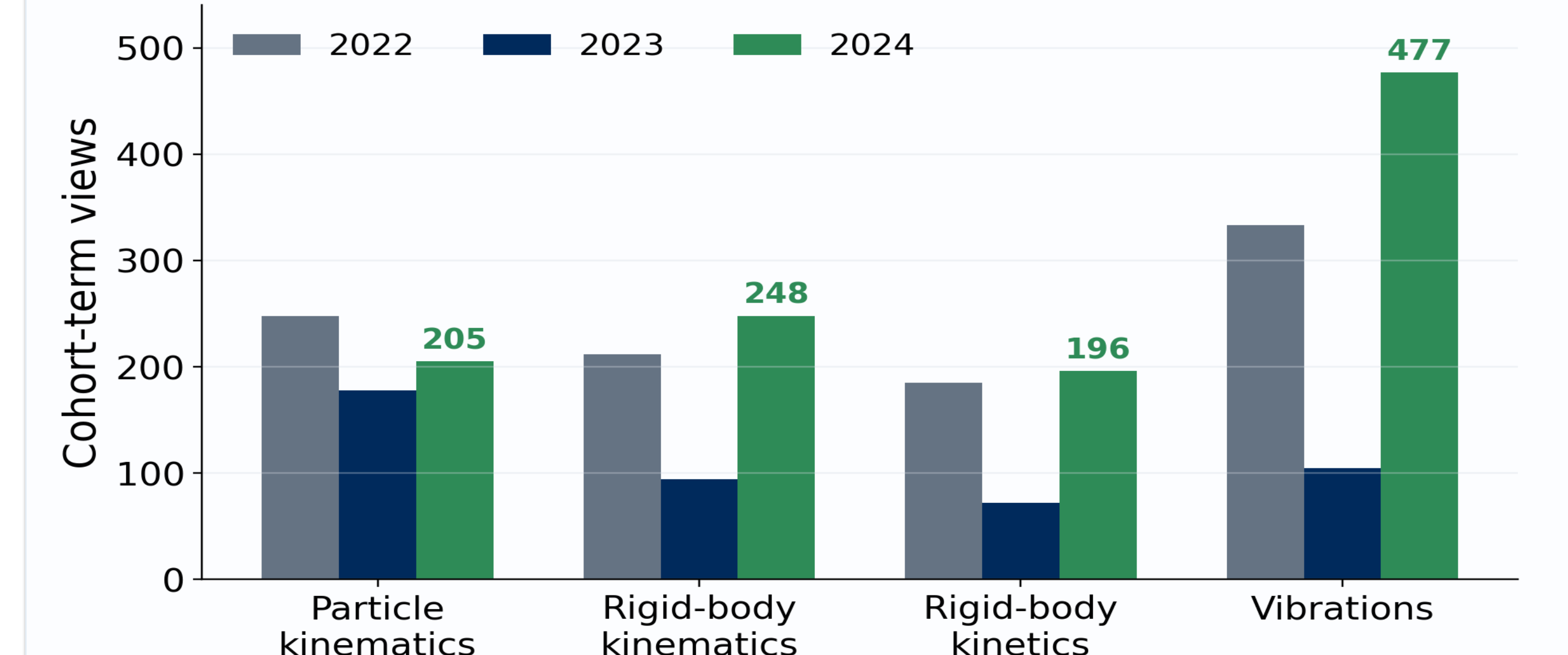
SVD/PCA flagged 2023 structural fragmentation



PC1 variance dropped in the 2023 final (34.6%) and rebounded in 2024 (41.5%), consistent with a more fragmented performance structure during the dip.

7. Resource activation

Summary resources were re-activated in 2024



Static summary resources were used much more in 2024: Vibrations rose from 105 to 477 cohort-term views.

8. Recommendations

- Flag topics with negative late-topic gaps.
- Track thread depth, not just answer counts.
- Target SVD/PCA bottlenecks with worked examples and office hours.
- Re-check after each support cycle.

Conclusion

Routine assessment and forum analytics can serve as a practical diagnostic instrument for allocating limited instructional support in large-enrollment gatekeeper courses.